Troy Mayer

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EDUCATION

Purdue University, School of Mechanical Engineering and School of Science **West Lafayette, IN**
B.S. Mechanical Engineering; B.S. Applied Statistics; Minor in Economics; Degree GPA 3.4/4.0 **May 2028**

PROFESSIONAL EXPERIENCE

Boston Dynamics **Waltham, MA**
Sensors and Perception Co-Op – Atlas | *Python, MuJoCo, Flask, SQLite, CATIA, Linux* **July 2026 – Present**

- Built a 3D impact-energy model using Khatib's operational-space formulation, estimating each fall strike's energy from logged joint states, the mass matrix and reflected rotor inertia, verified against hand-derived Jacobians.
- Designing CATIA fixtures and mounts, led by drop rigs that validate it: a rail-guided die spring on a two-axis gimbal centered at the contact point, aimed along each impact so its compression reads the full 3D energy.
- Built an impact-forensics ETL pipeline and Flask/Plotly dashboard that loads ~900 robot test logs into SQLite, giving engineers a 3D heatmap of ~28k impacts on the robot's mesh and per-event waveforms during live testing.
- Modeled hardware thermal behavior in Python, fitting heating curves with Savitzky-Golay filtering and multi-start least squares to predict steady-state temperatures across hundreds of units without full-length tests.

Chortos Lab – Soft Adaptive Systems and Structures **West Lafayette, IN**
Undergraduate Researcher | *Python, LabVIEW, MATLAB, Fusion 360* **August 2025 – May 2026**

- Designed an experiment to characterize the stress–strain response of ionic dielectric gels under voltage, building 15 Fusion 360 fixtures that hold every sample in the same position so formulations compare directly.
- Automated LabVIEW and MATLAB acquisition from a Keyence laser displacement head via an NI USB-6001, and wrote a Python computer-vision tracker measuring cantilever deflection to 10 µm against an in-frame micrometer scale.
- Formulated, synthesized and direct-ink-write printed gel variants, using those tests to tune their chemistry until electromechanical force output rose by over 50%.

Emerson – ASCO **Florham Park, NJ**
Mechanical and Manufacturing Engineering Intern | *Creo, CNC, GD&T* **May 2025 – August 2025**

- Designed 30+ rotary valve components in Creo, verified by hand stress analysis, and machined pistons to ±0.05 mm.
- Root-caused actuator deadband from cycle-test data and wrote an assembly procedure that cut it by over 10%.

PROJECTS

Purdue ASME – Biomechanics **West Lafayette, IN**
Active Orthosis – Electronics Team Lead | *C++17, FreeRTOS, KiCad, GoogleTest* **April 2024 – May 2026**

- Designed a 1 kHz feed-forward PI velocity controller with anti-windup in C++17 on an ESP32-S3, tuned from a first-order motor model to hold simulated speed error to 0.8%, guarded by 10 latching fault monitors and a watchdog.
- Cut EMG-to-motion latency from 103 to 43 ms with a Bayesian estimator, simulated on 24 real recorded contractions.
- Led a 5-engineer team redesigning the 4-channel EMG board in KiCad with an INA333 instrumentation front end, right-leg drive and ISO7763 motor isolation, raising day-to-day gesture decoding from 31% to 87% in emulated tests.

Asteroid Hopping Robot – Independent Research
Purdue University & Universidad Carlos III de Madrid | *Fusion 360, 3D Printing* **October 2026 – Present**

- Designed the full robot as a parametric Fusion 360 assembly: three five-bar legs driven by GM2804 gimbal motors through 3:1 timing belts, two internal reaction wheels, and an XL330 servo arm with a differential wrist and gripper.
- Building it to reshape its legs between hops, with push-off control optimized for a target launch speed, leg impedance control to absorb landings and reaction-wheel control to reorient mid-flight.
- Developing a pendulum landing rig and risk-limited hop policies that trade distance against escape and tip-over.

Legged Robot State Estimation and Control
Personal Project | *C++, Eigen, MuJoCo, GoogleTest, Python* **August 2026 – October 2026**

- Built a contact-aided error-state Kalman filter in C++/Eigen that fuses a simulated quadruped's 1 kHz IMU with leg kinematics, estimating base velocity to 0.013 m/s RMSE against 1.38 m/s for IMU-only.
- Verified every Jacobian against finite differences, then ran a Raibert trot controller and a QP balance controller on the estimate instead of ground truth, walking 5 seeds with zero falls and velocity tracking within 0.004 m/s.

Caterpillar (via Purdue Data Mine) **West Lafayette, IN**
Machine Learning Researcher | *Python, Snowflake, Power BI* **January 2026 – May 2026**

- Consolidated five years of noisy paint-line data and built LSTM and XGBoost forecasts of primer, topcoat and solvent use, contributing to a 4.7% cut in material usage and a 3.3% cut in cycle time.

SKILLS & INTERESTS

- **Tools:** C++, Python, MATLAB, SQL, LabVIEW, Eigen, MuJoCo, FreeRTOS, GoogleTest, KiCad, CATIA, Creo, Linux
- **Languages:** English (Native), Turkish (Fluent), Spanish (Fluent - New Jersey Seal of Biliteracy)
- **Clubs & Hobbies:** Club Swimming, House Band, Lifting, Hiking, Guitar, Saxophone